



BASE24 IMNi / IMNsc

IMNA USER GUIDE

Version 2.2

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REVISION HISTORY

Version	Date	Author	Comments
1.1		ACI	- Processing code now expects text codes instead of numeric ISO codes (Infoman # 1070) - Msg Type is now a required field (Infoman #1299) - Response Code has been removed as it is not used by the IMNA screen (Infoman #1260)
1.2		ACI	- The capture date is edit-checked to ensure the entered month/day is valid. - A screen freeze when switching logical networks has been fixed. - Edits have been added to record create date/time and local terminal date/time to ensure valid entry of these fields. Date format is now YY/MM/DD. Time format is now HH:MM:SS. (Infoman #1300) - Source node id is no longer an input field. The source node is obtained from the IMNM process of the logical network and displayed on the screen. (Infoman #1302) - Record creation date/time fields have been moved to the top of the screen. If these fields are changed to a non-blank value prior to edit/send then an error message is displayed. (Infoman #1259) - Support has been added for the F2 function key to read a specific record from the IMNADMN file based on a user entered record create date/time. (Infoman #1262) - The destination IMNi/IMNsc process is now supplied by the IMNM instead of being entered by the user. The destination and msg class are edit-checked by the IMNM. If there is no IMNi/IMNsc process available that is not in SAF forward only mode then an error message is displayed in response to the user pressing the F7 send key. (Infoman #1458) - When scanning through the IMNADMN file using the F6 key, records may be filtered by node id so that only those records with source, issuer or acquirer nodes matching a specified node will be retrieved. (Infoman #1458)
1.3	Jan. 4, 1991	ACI	- The capture date is optional. If omitted IMNA will request the current switch date from IMNM when performing the edit (F1) or send (F7) functions. - PAN and issuer node are now cross-checked. The

			issuer node may be omitted if a PAN is provided and IMNA will fill in the issuer node if the PAN has a valid prefix. If the issuer node is provided, it is checked against the PAN if any. The PAN is required for 620 and 622 type requests (Message Type 3). • The response code returned from IMNi/IMNsc is displayed on the screen.
1.4	Jul. 20, 1999	ACI	• This document is updated to include Shared Cash Processing Codes.
2.0	June 19, 2000	ACI	• Updates to identify mandatory versus optional input fields on IMNA Screen.
2.1	April 26, 2004	ACI	• Introduce two new fields (Alias/Password) on the screen.
2.2	August 2, 2007	ACI	• Update Appendix A with new IMNi Processing Codes for iDebit

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1. INTRODUCTION

1.1 About This Guide

This guide provides detailed information on the operation of the BASE24 IMN Administrative (IMNA) Screen used in the Interac IDP (Interac Direct Payment) and SCD (Shared Cash Dispensing) environments.

It explains the use of the IMNA screen for the generation of administrative messages for transaction tracing and adjustment and text transmission between nodes. It also explains how to access the BASE24 IMNADMN file from this screen.

1.2 The Main Users of This Guide

The main users of this guide will be Card Administrative staff responsible for handling transaction adjustments and transaction traces.

1.3 Related Documentation

- ACI FIL External Design Document
- BASE24 Network Control Operators Manual
- INTERAC EFT/POS Financial Institution Layer Functional Specifications Version 3.0

1.4 Topics in This Guide

The following topics are covered in this guide:

Screen Fields A summary of all data fields, including acceptable values and edit checks performed.

Function Keys A brief description of the functions available on this screen.

How To Use This Screen
A description of how to generate administrative transactions and access the IMNADMN file.

2. SCREEN FIELDS

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BASE24-SWI    IMN ADMIN MESSAGES      IART          99/07/02  16:23  01 OF 01
RECORD CREATION: (MM DD YY)          TIME:          SEARCH NODE:
MSG TYPE: 0    (0=TEXT; 1 = TRACE REQUEST; 2 = TRACE RESPONSE; 3 = ADJUST)
          SOURCE NODE: 444444 ACQ NODE          ISS NODE          MSG-CLASS
SCD TRANSACTION (Y/N):
PAN:                                TERMINAL LOCAL DATE(MM DD YY):
TRXN AMOUNT:                        TERMINAL LOCAL TIME(HH MM SS):
PROCESSING CODE:                    TERMINAL ID:
ORIG SETTLE DATE (MM DD):          TERMINAL OWNER ID:
REFERENCE/SYSTEM TRACE NUMBER:      TRXN FEE:
ALIAS:                                PASSWORD:
                                     TRACE ACTION/REASON CODE:

TEXT:

DESTINATION IMNI:                    RESPONSE CODE:

***** BASE24 *****
          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F7-SEND TO NET    SF7-CONFIRM    F12-HELP    F16-EXIT

```

Record Creation/Time

This is the key field for the IMNADMN file. It is used for record retrieval only, in which case it requires manual input of a valid date and time (MM DD YY HH MM SS CC). This marks the timestamp of the administrative transaction record to be retrieved.

Search Node

This is used as a filter for searching through the IMNADMN file using the 'F6-Read Next Record' key. Only those records with a node ID (source, issuer or acquirer) matching the search node will be retrieved. A blank value indicates the node ID matching criterion is not in effect and all records will be retrieved.

Msg Type

This field is used to determine the type of message to be sent. Valid types are as follows:

Text	0
Trace Request	1
Trace Response	2
Adjust	3

It is a required field for function key 'F7-Send To Net'.

Source Node

This field is the IMN Node Number of the Source node of this transaction. It is returned by the IMNM process of the

current logical network. It is for information purposes only.

Acq Node

This field is the IMN Node Number of the Acquirer node for this transaction. It is a required numeric field for function key 'F7-Send To Net'.

Iss Node

This field is the IMN Node Number of the Issuer node for this transaction.

PAN

This field is the Primary Account Number of the cardholder whose transaction is being traced or adjusted. If entered, it must be numeric, left justified and blank filled with no embedded blanks.

Notes on Iss Node and PAN:

1. For 'F1-Edit Data' and 'F7-Send To Net' function keys:

If Msg Type is 0, 1, or 2

- If the PAN is entered, IMNA looks up the PAN prefix and fills in the Issuer node.
- If the PAN is not entered, the Issuer node is required as a numeric field.

If Msg Type is 3

- The Issuer node is not required, however the PAN is.

If both the Issuer node and PAN are entered, the PAN is checked to ensure that it is valid for the Issuer node entered.

2. Either Acq Node or Iss Node must match Source Node. Acq Node cannot be the same as Iss Node.

Msg Class

This field is the message class to be used for this transaction. It is a required numeric field and validated by IMNM.

SCD Transaction

This field is used to specify whether or not the current transaction is of type SCD. Valid values are 'Y' or 'N'. If the processing code refers to a SCD transaction type, this field will be defaulted to 'Y' and protected.

Terminal Local Date

This field is the date at the POS terminal of the transaction being traced or adjusted. It must be a valid date in the format MM DD YY. It is a required field for all available message types.

Trxn Amount	This field is the adjustment amount for adjustment transactions. It must be a numeric value with no more than two decimal places. It is a required field when Msg Type is 3. The value of the field must be negative for credit adjustment transactions.
Terminal Local Time	This field is the time at the POS terminal of the transaction being traced or adjusted. It must be a valid time in the format HH MM SS. It is a required field for all available message types.
Processing Code	This field indicates the type of transaction to be traced or adjusted. Please refer to Appendix A – Processing Codes, for the list of possible values.
Terminal ID	This field is the retailer assigned terminal ID from the transaction being traced or adjusted. It is an optional field and is not validated.
Orig Settle Date	This field is the settlement date for the original transaction. The format is MM DD and it must be a valid month and day. It is a required field and validated by IMNM.
Terminal owner ID	This field is the terminal owner ID from the transaction being traced or adjusted. It is an optional field and not validated.
Reference/System Trace Number	This field is the retailer-assigned transaction reference number. It is not edit-checked if the ‘SCD Transaction’ flag is set to ‘Y’. It is an optional field.
Trxn Fee	This field is reserved for future use. Any value entered will have no effect on the transaction being sent.
Alias	This field is a user name that is associated with a uniquely identified user in order to gain access to the NET24-XPNET command. It is an optional field when the security of server NCPI is disabled. If the security of server NCPI is enabled, this field will be verified by the NET24-XPNET. For more information on how to configure security parameter of server NCPI to be enabled (ENABLE-SECURITY), please reference to the NET24-XPNET Network Control Operations Guide.

Password	This field is a password associated with the specified alias. It is an optional field when the security of server NCPI is disabled. If the security of server NCPI is enabled, this field will be verified by the NET24-XPNET. Passwords are case sensitive. For more information on how to configure security parameter of server NCPI to be enabled (ENABLE-SECURITY), please reference to the NET24-XPNET Network Control Operations Guide.
Trace Action/Reason Code	This field is the Trace Action (for trace responses) or Reason Code (for trace requests). Please refer to Appendix B – Trace Action/Reason Codes, for the list of possible values.
Text	This field is a free format text field for comments/instructions regarding the transaction. It must not contain any semi-colons.
Destination IMNI	This field is the symbolic name of the BASE24 IMNi/IMNsc process to which the administrative transaction will be sent. It is set by IMNM when the 'F1-Edit' or the 'F7-Send To Net' key is pressed. It is for information purposes only.
Response Code	This field shows the response code returned from the IMNi/IMNsc process. It is for display purposes only and is set to blanks when the 'F1-Edit' or 'F7-Send To Net' key is pressed. A value of "00" indicates that the message was successfully sent to the destination node. Any other value indicates an error. Refer to the EFT/POS FIL Functional Specifications for translation of the codes.
File Destination	This field is used in conjunction with the 'F16-Exit' function key to switch to another BASE24 screen. It must contain a valid BASE24 screen name.
New Logical Network ID	This field is used in conjunction with the 'F13-New Logical Net' function key to change BASE24 Logical Networks. It must contain a valid BASE24 Logical Network name.

3. FUNCTION KEYS

F1-Edit Data	This function edit-checks the IMNA screen fields, one by one, to determine if a value was entered (for mandatory input fields) or, if entered, is valid.
F2-Read Record	This function retrieves the IMNADMN record that matches the Record Creation date and time entered by the user.
F4-Delete Record	This function deletes the record with the specified Record Creation date and time from the IMNADMN file.
F6-Read Next Record	This function retrieves the next record with a Record Creation time or date greater than that specified on the screen. If a non-blank value is entered in the Search Node field, the IMNADMN file is scanned until a record with a node id (either source, issuer or acquirer) that matches the Search Node is found (or end of file is reached).
F7-Send To Net	This function validates the transaction on the IMNA screen, and if it is correct, sends it to the BASE24 IMNi/IMNsc selected by the IMNM process (see 'F1-Edit Data' above).
SF7-Confirm Send	This function confirms the execution of 'F7-Send To Net'.
F8-Clear Screen	This function replaces all screen fields with blanks.
F10-Print Screen	This function prints the screen to the print destination specified at logon.
F12-Display Help Screen	This function displays the IMNA Help screen.
F13-Change Current Logical Net	This function switches the current BASE24 logical net to the specified one. This function should not be used except under direction from the system manager.
F16-Exit Or Go To File	This function takes the user to the BASE24 screen specified in the File Destination field, or to the Virtual Menu if no destination is selected.
SF16-Logoff	This function logs off the user from the BASE24 system.

4. HOW TO USE THIS SCREEN

4.1 Accessing The IMNA Screen

The IMNA screen can be reached directly from the BASE24 Logon screen by typing IMNA in the File Destination field before logging on to the BASE24 system. To reach the IMNA screen from any other BASE24 screen, type IMNA in the File Destination field and press F16. The IMNM process for the logical network must be running in order for the source id to be set on the screen and for editing of the transaction ('F1-Edit Data' or 'F7-Send To Net') to be successful.

4.2 Accessing The IMNADMN File

The IMNADMN file is used for storing all administrative transactions sent to a node. All valid ISO 06XX messages are logged to this file. Transaction records are keyed by creation date and time. These records can be displayed on the IMNA screen.

To retrieve a record, enter the creation date and time in the Record Creation field at the top of the screen and press 'F2-Read Record' or 'F6-Read Next Record'. The 'F2-Read Record' key will retrieve only the record with the Record Creation date/time specified or return an error. The 'F6-Read Next Record' key will retrieve the next record in the file with a Record Creation date/time greater than the one specified. If the Search Node field is non-blank, the 'F6-Read Next Record' key will retrieve the next record with a node ID (one of source, issuer or acquirer) that matches the Search Node.

To see more records, in order of their receipt, press 'F6-Read Next Record' until the desired record is displayed. Note that the Search Node field will filter responses if it is non-blank.

To delete a record from the IMNADMN file, enter the record's creation date and time and press 'F4-Delete Record'. When the record has been successfully deleted, a message to that effect is displayed at the bottom of the IMNA screen. An error message is displayed if the record does not exist or if an error occurred.

4.3 Generating Transactions

The type of administrative transaction sent depends upon the combination of Message Type, Acquirer Node, and Source Node on the IMNA screen. The combination of values and their corresponding ISO Message types are listed below:

ISO Type	Msg Type	Node Restrictions
0600 - Administrative Request (Acquirer)	0,1,2	Acq Node = Source Node
0602 - Administrative Request (Issuer)	0,1,2	Iss Node = Source Node
0620 - Administrative Adjustment (Acquirer)	3	Acq Node = Source Node
0622 - Administrative Adjustment (Issuer)	3	Iss Node = Source Node

The IMN Node on which the IMNA is running must have an available connection to the Acquirer or Issuer node specified (depending on the ISO Msg Type sent) with the specified Msg Class in order to send administrative transactions to that node.

To generate an administrative transaction, enter all required data on the IMNA screen and press 'F7-Send to Net'. The data will be validated, and if no errors are found, the transaction is sent to the IMNi/IMNsc selected by the IMNM. If there is an error or the message cannot be sent, an error message is displayed at the bottom of the screen, indicating the type of error that occurred.

The following table identifies the IMNA screen fields that require input (i.e. 'Y') when using function key 'F7-Send To Net'.

Msg Type Screen Fields	0-Text	1-Trace Request	2-Trace Response	3-Adjust
Record Creation				
Time				
Search Node				
Msg Type	Y	Y	Y	Y
Source Code				
Acq Node	Y	Y	Y	Y
Iss Node/PAN	Either required	Either required	Either required	Only PAN required
Msg Class	Y	Y	Y	Y
SCD Transaction				
Terminal Local Date	Y	Y	Y	Y
Trxn Amount				Y
Terminal Local Time	Y	Y	Y	Y
Processing Code	Y	Y	Y	Y

Msg Type Screen Fields	0-Text	1-Trace Request	2-Trace Response	3-Adjust
Terminal ID				
Orig Settle Date	Y	Y	Y	Y
Terminal Owner ID				
Reference Number/System Trace Number				
Trxn Fee				
Alias	(1)	(1)	(1)	(1)
Password	(1)	(1)	(1)	(1)
Trace Action/Reason Code		Y	Y	Y
Text	Y			
Destination IMNI				
Response Code				

(1) For more details, please refer to Section 2 SCREEN FIELDS

APPENDIX A: PROCESSING CODES

The text Processing Code field on the IMNA screen is converted by the IMNA requester to a numeric ISO Processing Code that is used by the IMNi/IMNsc to build the 06XX ISO message. This ISO Code is converted by the receiving IMNi/IMNsc to a BASE24 Processing Code using the BASE24 ATM/POS Transaction Type Table shown below.

IMNi

Text Code	Transaction Type	ISO Processing Code	BASE24 Processing Code
PUR-SAV	Purchase from savings	001000	102110
PUR-CHQ	Purchase from chequing	002000	102010
PAP-SAV	Pre-authorization - savings	031000	112110
PAP-CHQ	Pre-authorization - chequing	032000	112010
PAC-SAV	Completion - savings	171000	122110
PAC-CHQ	Completion - chequing	172000	122010
REF-SAV	Return/Adjustment - savings	200010	142110
REF-CHQ	Return/Adjustment - chequing	200020	142010
RFR-SAV	Debit reversal - savings	021000	222110
RFR-CHQ	Debit reversal - chequing	022000	222010
PRR-SAV	Credit reversal - savings	220010	212110
PRR-CHQ	Credit reversal - chequing	220020	212010
FAI-SAV	Balance inquiry - savings	301000	172110
FAI-CHQ	Balance inquiry - chequing	302000	172010
Ref-IDB	Return/Adjustment – Idebit/IOP	200050	142500
Pur-IDB	Purchase from Idebit/IOP	005000	102500
NO-CODE	Text message	No processing code sent in ISO message	

IMNsc

Text Code	Transaction Type	ISO Processing Code	BASE24 Processing Code
WTH-SAV	Withdrawal from savings	011000	101100
WTH-CHK	Withdrawal from chequing	012000	100100
CSH-CRE	Withdrawal from credit card	013000	103100
NO-CODE	Text message	No processing code sent in ISO message	

APPENDIX B: TRACE ACTION/REASON CODES**Trace Action Codes**

<u>Code</u>	<u>Description</u>
01	Being investigated
02	Adjustment will follow
03	Transaction is valid
04	Adjustment applied

Trace Reason Codes

<u>Code</u>	<u>Description</u>
01	Customer inquiry
02	Customer dispute
03	Fraudulent transaction
04	Customer denies txn finalized
06	Transaction failed edit check
12	Invalid transaction
13	Invalid amount
14	Invalid card number
23	Unacceptable transaction fee
33	Expired card
36	Restricted card
39	No credit account
41	Lost Card
43	Stolen card
51	Non-sufficient funds
52	No chequing account
53	No savings account
54	Expired card
55	Incorrect PIN
57	Trxn not permitted to cardholder
61	Exceeds purchase amount limit
62	Card temporarily disallowed
65	Exceeds purchase frequency limit
77	Invalid capture date
78	Amount error
79	Time expired
94	Duplicate Transmission

APPENDIX C: IMNA – ISO FIELD MAPPING

IMNA Field	ISO Field	Bitmap Number
Acq Node	Acquiring Institution ID Code	32,48
Iss Node	Issuing Institution ID Code	48
PAN	Primary Account Number	2
Trxn Amount	Transaction Amount	4
Processing Code	Processing Code	3
Orig Settle Date	Capture Date	17
Reference Number	Retrieval Reference Number	37
Terminal Local Date	Local Transaction Date	13
Terminal Local Time	Local Transaction Time	12
Terminal ID	Card Acceptor Terminal ID	41
Terminal Owner ID	Card Acceptor ID Code	42
Trace Action/Reason Code	Admin Data Trace Action/Reason Code in Addition Data Private	48
Source Node	Destination ID in Additional Data Private	48
Text	Admin Message in Additional Data Private	48

Note: The ISO Admin Type in the Additional Data Private field is determined by the values of the Msg Type and Trxn Amount fields as follows:

Admin Type	Msg Type	Trxn Amount
01	0	
10	1	
12	2	
14	3	Trxn Amount >= 0
16	3	Trxn Amount < 0